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1) Correlation & Covariance

X	Y
1	2
2	4
3	5
4	4
5	5

X: Hours Studied
Y: Exam scores

Compute:-
1) Covariance
2) Correlation

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i) Correlation

$$r = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

ii) $$Cov(X,Y) = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

iii) $$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})^2}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x^{(i)}$$

$\bar{x} = 3$
$\bar{y} = 4.$

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})^2}$$

$(x-\bar{x})$	x	$(x-\bar{x})^2$	y	$(y-\bar{y})^2$	$(y-\bar{y})$
-2	1	4	2	4	-2
-1	2	1	4	0	0
0	3	0	5	1	1
1	4	1	4	0	0
2	5	4	5	1	1

$$\sigma_x = \sqrt{\frac{1}{5} \times 10}$$
$$= \sqrt{2}$$
$$= 1.414$$

$$\sigma_y = \sqrt{\frac{1}{5} \times 6}$$
$$= \sqrt{1.2} = 1.095$$

$$\mathrm{cov}(x,y) = \frac{1}{5} \times 6 = 1.2$$

$$r = \frac{1.2}{1.548} = 0.775$$

Compute the eigenvalues & eigenvectors of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)$$

$$= \det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right)$$

$$\det\begin{pmatrix} 2-\lambda & 1-0 \\ 1-0 & 2-\lambda \end{pmatrix}$$

$$\Rightarrow (2-\lambda) \times (2-\lambda) - (1)(1)$$

$$\Rightarrow 4 - 2\lambda - 2\lambda + \lambda^2 - 1$$

$$\Rightarrow \lambda^2 - 4\lambda + 3$$

$$\Rightarrow (\lambda - 3)(\lambda - 1)$$

$$\lambda = 3, \quad \lambda = 1.$$

eigenvector

$(A - \lambda I) V_0$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\therefore \begin{bmatrix} -x_1 + x_2 \\ x_1 - x_2 \end{bmatrix} = 0$$

$$-x_1 + x_2 = 0$$

$$\text{or } x_1 - x_2 = 0$$

$$x_1 = x_2$$

for $\lambda = 1$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x_1 + x_2 \\ x_1 + x_2 \end{bmatrix} = 0$$

$$x_1 + x_2 = 0$$